Course Code	21CSC203P	Course Name	ADVANCED PROGRAMMING PRACTICE	Course Category	C	PROFESSIONAL CORE	L	T	P	C
							3	1	0	4

Pre-requisite Courses	Nil	Co- requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	School of Computing	Data Book / Codes / Standards	Nil		

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	understand the paradigm functionalities and their hierarchy
CLR-2:	deploy structural, procedural, and Object-Oriented Programming Paradigm
CLR-3:	demonstrate the event, Graphical User Interface, and declarative Paradigm with a java application
CLR-4:	extended knowledge on logic, functional, network and concurrent Paradigm
CLR-5:	symbolic, Automata-based, and Event with a python application

Course Outcomes (CO):	At the end of this course, learners will be able to:	Program Outcomes (PO)												Program Specific Outcomes		
		1	2	3	4	5	6	7	8	9	10	11	12			
		Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CO-1:	devise solutions to the various programming paradigm	3	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO-2:	express proficiency in the usage of structural, procedural, and Object-Oriented Program	3	2	-	1	-	-	-	-	-	-	-	-	2	-	-
CO-3:	determine the Java application using declarative, event, and graphical user interface paradigm	3	-	2	1	2	-	-	-	1	-	-	-	2	-	-
CO-4:	express proficiency in the usage of logic, functional, network, and concurrent Paradigm	3	2	-	1	-	-	-	-	-	-	-	-	2	-	-
CO-5:	determine the Python application using symbolic, automata-based, and graphical user interface programming paradigms	3	-	2	1	2	-	-	-	1	-	-	-	2	-	-

Unit-1 - Introduction to Programming Paradigm — **12 Hour**

Programming Languages – Elements of Programming languages - Programming Language Theory - Bohm- Jacopini structured program theorem - Multiple Programming Paradigm – Programming Paradigm hierarchy – Imperative Paradigm: Procedural, Object-Oriented and Parallel processing – Declarative programming paradigm: Logic, Functional and Database processing - Machine Codes – Procedural and Object-Oriented Programming – Suitability of Multiple paradigms in the programming language - Subroutine, method call overhead and Dynamic memory allocation for message and object storage - Dynamically dispatched message calls and direct procedure call overheads – Object Serialization – parallel Computing

Unit-2 - Java Programming Paradigms — **12 Hour**

Object and Classes; Constructor; Data types; Variables; Modifier and Operators - Structural Programming Paradigm: Branching, Iteration, Decision making, and Arrays - Procedural Programming Paradigm: Characteristics; Function Definition; Function Declaration and Calling; Function Arguments - Object-Oriented Programming Paradigm: Abstraction; Encapsulation; Inheritance; Polymorphism; Overriding - Interfaces: Declaring, implementing; Extended and Tagging - Package: Package Creation.

Unit-3 - Advanced Java Programming Paradigms — **12 Hour**

Concurrent Programming Paradigm: Multithreading and Multitasking; Thread classes and methods - Declarative Programming Paradigm: Java Database Connectivity (JDBC); Connectivity with MySQL – Query Execution; - Graphical User Interface Based Programming Paradigm: Java Applet: Basics and Java Swing: Model View Controller (MVC) and Widgets; Develop a java project dissertation based on the programming paradigm.

Unit-4 - Pythonic Programming Paradigm — **12 Hour**

Functional Programming Paradigm: Concepts; Pure Function and Built-in Higher-Order Functions; Logic Programming Paradigm: Structures, Logic, and Control; Parallel Programming Paradigm: Shared and Distributed memory; Multi-Processing – Ipython; Network Programming Paradigm: Socket; Socket Types; Creation and Configuration of Sockets in TCP / UDP – Client / Server Model.

Unit-5 - Formal and Symbolic Programming Paradigm	***12 Hour***
Automata Based programming Paradigm: Finite Automata – DFA and NFA; Implementing using Automaton Library - Symbolic Programming Paradigm: Algebraic manipulations and calculus; Sympy Library - Event Programming Paradigm: Event Handler; Trigger functions and Events – Tkinter Library. Develop a python-based project dissertation based on the programming paradigm.	

Learning Resources	1. *Elad Shalom, A Review of Programming Paradigms throughout the History: With a suggestion Toward a Future Approach, Kindle Edition, 2018* 2. *Maurizio Gabbrielli, Simone Martini, Programming Languages: Principles and Paradigms, 2010.*	3. *Herbert Schildt, Java: The Complete Reference Seventh Edition, 2016.* 4. *Mark Lutz, Programming Python: Powerful Object-Oriented Programming, 2011.*

Learning Assessment

	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)						Final Examination (0% weightage)	
		Formative CLA-1 Average of unit test (20%)		Project Based Learning CLA-2 (60%)		Report and Viva Voce (20% weightage)			
		Theory	Practice	Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	30%	-	--	20%	-	10%	-	-
Level 2	Understand	30%	-	-	20%	-	10%	-	-
Level 3	Apply	20%	-	-	20%	-	10%	-	-
Level 4	Analyze	20%	-	-	20%	-	10%	-	-
Level 5	Evaluate	-	-	-	10%	-	30%	-	-
Level 6	Create	-	-	-	10%	-	30%	-	-
	Total	100 %		100 %		100 %		-	

Course Designers

Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1. *Mr. N. Venkatesh, Tech Lead, Honeywell, Bengaluru, Karnataka, India*	1. *Dr. Sudeepta Mishra, Assistant Professor, Computer Science and Engineering, Indian Institute of Information Technology, Ropar, Punjab.*	1. *Dr Ramkumar J, SRMIST*